

Name _____ Teacher _____



GOSFORD HIGH SCHOOL

2015

HIGHER SCHOOL CERTIFICATE

ASSESSMENT TASK 1

MATHEMATICS – EXTENSION 1

Time Allowed - 60 minutes plus 5 minutes reading time

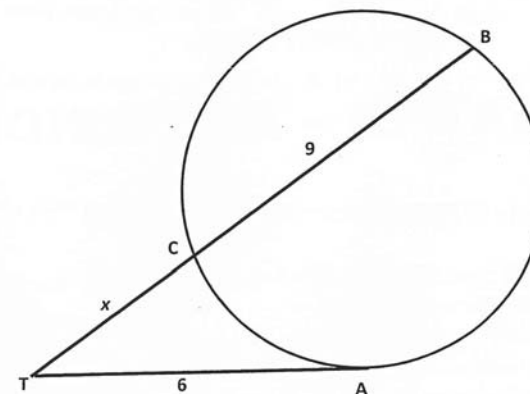
- Write using a black or blue pen. Black pen is preferred.
- Board approved calculators may be used.
- Answers to the multiple choice are to be done on the answer sheet provided.
- Start each section on a **new** page. **Each section** will be collected **separately**.
- In Sections 2-4, show relevant mathematical reasoning and/or calculations.

Section 1 Multiple choice	4 questions worth 1 mark each.	/4
Section 2	Parametric treatment of the parabola	/14
Section 3	Polynomials	/16
Section 4	Miscellaneous review	/12
TOTAL		/46

Section 1 Objective response questions (1 mark each)

Answer each question on the multiple choice answer sheet provided.

- 1) What is the sum of the zeros of $(x) = 2x^3 + 8x^2 - x + 4$?
(A) 4 (B) -4 (C) 0 (D) -8
- 2) Line TA is a tangent to the circle at A and TB is a secant meeting the circle at B and C. Given that TA = 6, CB = 9 and TC = x, what is the value of x?



- (A) -12 (B) 2 (C) 3 (D) 4
- 3) The polynomial $P(x)$ is monic and of degree 5. It has a single zero at $x = -2$ and a double zero at $x = 1$ Its other two zeros are not real.
Which of the following equations **best** represents $P(x)$?
(A) $(x^2 - 1)(x + 2)(x - a)(x - b)$, where $b^2 - 4c < 0$
(B) $(x - 1)(x + 2)^2(x + a)(x + b)$, where $b^2 - 4c \geq 0$
(C) $(x + 2)(x - 1)^2(x^2 + bx + c)$, where $b^2 - 4c \geq 0$
(D) $(x + 2)(x - 1)^2(x^2 + bx + c)$, where $b^2 - 4c < 0$
 - 4) Evaluate

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 1}{7 + 4x^2}$$

- (A) $\frac{3}{7}$ (B) $\frac{3}{4}$ (C) $\frac{1}{7}$ (D) $\frac{1}{4}$

Section 2 Parametric treatment of the parabola (Start a new page)

MARKS

- 1) The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$
 - (a) Derive the equation of the tangent and the normal at the point P 3
 - (b) The tangent and the normal at P meet the axis of the parabola $x^2 = 4ay$ at the points T and N respectively. Show that $(TN)^2 = 4a^2(1 + p^2)^2$ 4
- 2) The variable point $(6t, 4t^2)$ lies on a parabola. Find the coordinates of the focus of this parabola. 2
- 3) Find the equation of the chord of contact of the tangents to the parabola $x^2 = 4y$ from the external point $(4, -6)$ 2
- 4) The points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on a parabola with focus S . PQ is a focal chord with equation $y - \frac{1}{2}(p + q)x + apq = 0$. Show that
 - (a) $pq = -1$ 1
 - (b) If $p + q = 4$ and using part (a), show that the midpoint of PQ is $(4a, 9a)$ 2

Section 3 Polynomials (Start a new page)

- 1) If α, β, γ are the roots of the cubic equation $x^3 - 3x + 2 = 0$ find
 - (i) $\alpha + \beta + \gamma$ 1
 - (ii) $\alpha\beta + \alpha\gamma + \beta\gamma$ 1
 - (iii) $\alpha\beta\gamma$ 1
 - (iv) $\alpha^2 + \beta^2 + \gamma^2$ 2
- 2) The polynomial $P(x) = x^3 - 6x^2 + kx + 14$ has a root at $x=1$.
 - (i) Determine the value of k 1
 - (ii) For this value of k , express $P(x)$ as a product of its linear factors 3
 - (iii) For this value of k , solve $P(x) > 0$ 1

- 3) When $P(x)$ is divided by $x^2 - 4$ the remainder is $2x + 3$. Find the remainder when $P(x)$ is divided by $x - 2$ 2
- 4) The equation $x^3 - mx + 2 = 0$ has two equal roots.
 - (i) Write down expressions for the sum of the roots (1 at a time) and the product of the roots. 2
 - (ii) Hence find the value of m 2

Section 4 Miscellaneous Topics (Start a new page)

- 1) Sketch showing all important features $y = \frac{x+1}{x^2+3x+2}$ 4
- 2) Using the graph from part 1) or otherwise, solve $\frac{x+1}{x^2+3x+2} > 0$ 2
- 3) The acute angle between the lines $y = (m + 2)x$ and $y = mx$ is 45° .
 - (i) Show that $\left| \frac{2}{m^2+2m+1} \right| = 1$ 1
 - (ii) Hence find any values of m 3
- 4) The point $(3, -4)$ divides the interval AB externally in the ratio $3:2$. If the coordinates of A are $(6, 5)$, then find the coordinates of B . 2

Section 1

$$1) -\frac{b}{a} = -\frac{-8}{2} \\ = -4$$

(B)

$$2) (9+x), x = 36 \\ x^2 + 9x - 36 = 0$$

$$(x+12)(x-3) = 0$$

$$x = -12 \text{ or } x = 3$$

can't be negative $\therefore x = 3$ only

(C)

$$3) (D)$$

$$4) \lim_{x \rightarrow \infty} \frac{\frac{3x^2}{x^2} + \frac{1}{x^2}}{\frac{7}{x^2} + \frac{4x^2}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{3 + \frac{1}{x^2}}{\frac{7}{x^2} + 4}$$

$$= \frac{3}{4} \quad (B)$$

Section 2

$$1) a) y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a} \\ = \frac{x}{2a}$$

at $x = 2ap$

$$m_T = \frac{2ap}{2a} \therefore m_N = -\frac{1}{p} \\ = p \quad \checkmark$$

$$\text{Equation of tangent is } y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2 \\ y = px - ap^2 \quad \checkmark$$

$$\text{Equation of tangent is } y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap \quad \checkmark$$

(b) Axis of parabola is $x = 0$

$$y = ap^2 \therefore T \text{ is } (0, -ap^2) \quad \checkmark$$

For normal

$$py - ap^3 = 2ap \\ py = 2ap + ap^3 \\ y = 2a + ap^2$$

$$\therefore N \text{ is } (0, 2a + ap^2) \quad \checkmark$$

$$\therefore (TN)^2 = (0-0)^2 + (2a + ap^2 + ap^2)^2 \quad \checkmark$$

$$= (2a + 2ap^2)^2$$

$$= (2a(1+p^2))^2$$

$$= 4a^2(1+p^2)^2 = 4a^2(1+p^2)^2 \quad \checkmark$$

$$2) x = 6t \rightarrow t = \frac{x}{6}$$

$$y = 4t^2 \therefore y = \frac{4x^2}{36}$$

$$y = \frac{x^2}{9}$$

$$\therefore x^2 = 9y \quad \checkmark \quad 4a = 9$$

$$a = \frac{9}{4}$$

$$\text{Focus is } (0, \frac{9}{4}) \quad \checkmark$$

$$3) \text{ Using } xx_1 = 2a(y+y_1) \quad \checkmark \text{ where } a = 1$$

$$4x = 2(y-6)$$

$$4x - 2y + 12 = 0$$

$$\therefore 2x - y + 6 = 0 \quad \checkmark$$

$$4(a) \text{ Using } y - \frac{1}{2}(p+q)x + apq = 0$$

Focal chord if passes through $(0, a)$

$$a + apq = 0$$

$$apq = -a$$

$$pq = -1 \quad \checkmark$$

$$(b) \text{ Midpoint } PQ \text{ is } \left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \right)$$

$$= \left(a(p+q), \frac{a(p^2+q^2)}{2} \right)$$

$$= \left(a(p+q), a \left[\frac{(p+q)^2 - 2pq}{2} \right] \right)$$

$$\text{but } p+q = 4 \text{ and } pq = -1$$

$$\therefore \text{midpoint } PQ = \left(4a, \frac{a(16+2)}{2} \right)$$

$$= (4a, 9a) \quad \checkmark$$

Section 3

$$x^3 + 0x^2 - 3x + 2 = 0$$

$$1) (i) -\frac{b}{a} = 0$$

$$(ii) \frac{c}{a} = -\frac{3}{1}$$

$$= -3$$

$$(iii) -\frac{d}{a} = -2$$

$$(iv) (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= \left(-\frac{b}{a} \right)^2 - 2\left(\frac{c}{a} \right)$$

$$= 0 - 2 \times -3$$

$$= 6$$

$$2) (i) 0 = 1 - 6 + k + 14$$

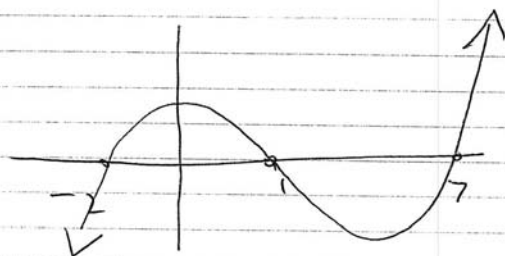
$$k = -9$$

$$(ii) P(x) = x^3 - 6x^2 - 9x + 14$$

$$\begin{array}{r}
 x^2 - 5x - 14 \\
 x-1 \overline{) x^3 - 6x^2 - 9x + 14} \\
 \underline{x^3 - x^2} \\
 5x^2 - 9x \\
 \underline{5x^2 + 5x} \\
 -14x + 14
 \end{array}$$

$$P(x) = (x-1)(x-7)(x+2)$$

(iii)



$$x > 7 \text{ or } -2 < x < 1$$

$$3) P(x) = (x^2 - 4) \cdot Q(x) + 2x + 3$$

$$P(2) = 2(2) + 3$$

$$= 7$$

$\therefore$ remainder is 7.

$$4) (i) x^2 B = -2 \quad 2x + B = 0$$

$$2x = -B$$

$$-2x = B$$

$$(ii) x^2(-2x) = -2$$

$$-2x^3 = -2$$

$$x^3 = 1$$

$$x = 1$$

$$\text{sub } x = 1$$

$$1 - m + 2 = 0$$

$$m = 3$$

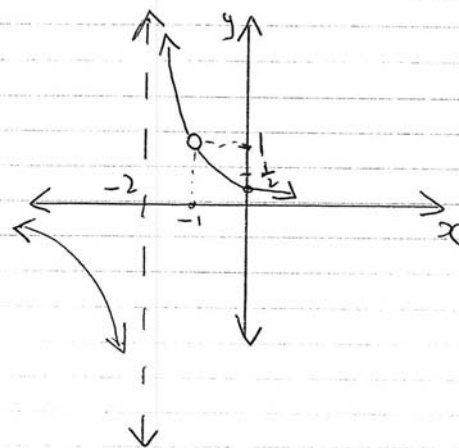
Section 4

$$1) y = \frac{x+1}{(x+2)(x+1)} \quad x \neq -2 \text{ or } -1$$

$$y = \frac{1}{x+2}$$

$$\lim_{x \rightarrow -1} \frac{1}{x+2}$$

$$= 1$$



$$2) x > -2, x \neq -1$$

$$3) (i) \tan 45^\circ = \left| \frac{m+2-m}{1+m(m+2)} \right|$$

$$1 = \left| \frac{2}{1+m^2+2m} \right|$$

$$(ii) \pm 1 = \frac{2}{1+m^2+2m}$$

$$m^2 + 2m + 1 = 2 \quad \text{OR} \quad -m^2 - 2m - 1 = 2$$

$$m^2 + 2m - 1 = 0 \text{ OR } m^2 + 2m + 3 = 0$$

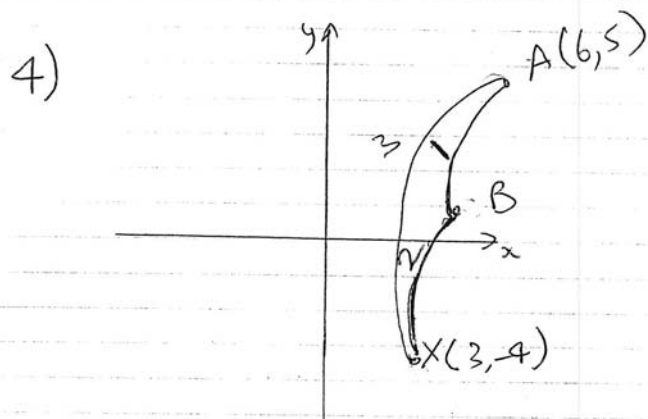
$$m = \frac{-2 \pm \sqrt{4 - 4 \times 1 \times -1}}{2} \quad (m+3)(m+1) = 0$$

$$\underline{m = -3 \text{ or } m = -1}$$

$$m = \frac{-2 \pm \sqrt{8}}{2}$$

$$m = \frac{-2 \pm 2\sqrt{2}}{2}$$

$$\underline{m = -1 \pm \sqrt{2}}$$



$$\begin{array}{ccc} A(6, 5) & & X(3, 4) \\ & \swarrow \quad \searrow & \\ & 1 \quad 2 & \\ & \vdots & \end{array}$$

$$\left(\frac{(2)(6) + (1)(3)}{3}, \frac{(2)(5) + (1)(4)}{3} \right)$$

B is (5, 2)